FINANCIAL STRATEGY ANALYSIS OF TESLA, INC.

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ABSTRACT

Financial Strategy Analysis of Tesla, Inc.

By

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With the continuous innovation of technology, human beings are paying more and more attention to the protection of the future earth's ecological environment. The development and use of new energy in various industries have gradually become a new competitive indicator. For the automotive industry, the gradual lack of petroleum resources and the harm of exhaust emissions to air pollution have made the development and utilization of new energy sources extremely urgent.

Tesla Motors was founded in 2003. It is a company dedicated to reducing the world's dependence on fossil fuel-powered vehicles by developing electric vehicles. For a long time, Tesla's future development is a topic that investors are keen on. As the first mover in the electric vehicle industry, Tesla has a lot to learn, and at the same time, there are many crises that need to be avoided. This research will analyze the financial situation of Tesla from the perspective of an investor to find out the factors of success and provide some suggestions for more investors.

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CHAPTER 1

Introduction

The increasingly harsh global ecological environment and climate change have become a severe problem. With the energy-saving and emission reduction of major automobile manufacturers, the research investment of large-displacement engines has become less and less. In the current environment of scarce oil resources, Mercedes-Benz and BMW have been trying to install the 2.0T engine on their flagship models to achieve small displacement and high horsepower. The German Karl Benz developed the world's first internal combustion engine in October 1885. But from the internal combustion engine to today's pure electric motor is less than 200 years old. Human beings are increasingly aware of the changes in the ecological environment. The development of electric vehicles has become an irreversible direction for the automotive industry. Tesla is currently the most popular new energy vehicle in the market. As a technology company that only produces electric vehicles, it is different from other hybrid manufacturers in the market. It is clear that Tesla has become a leader in the field of new energy vehicles.

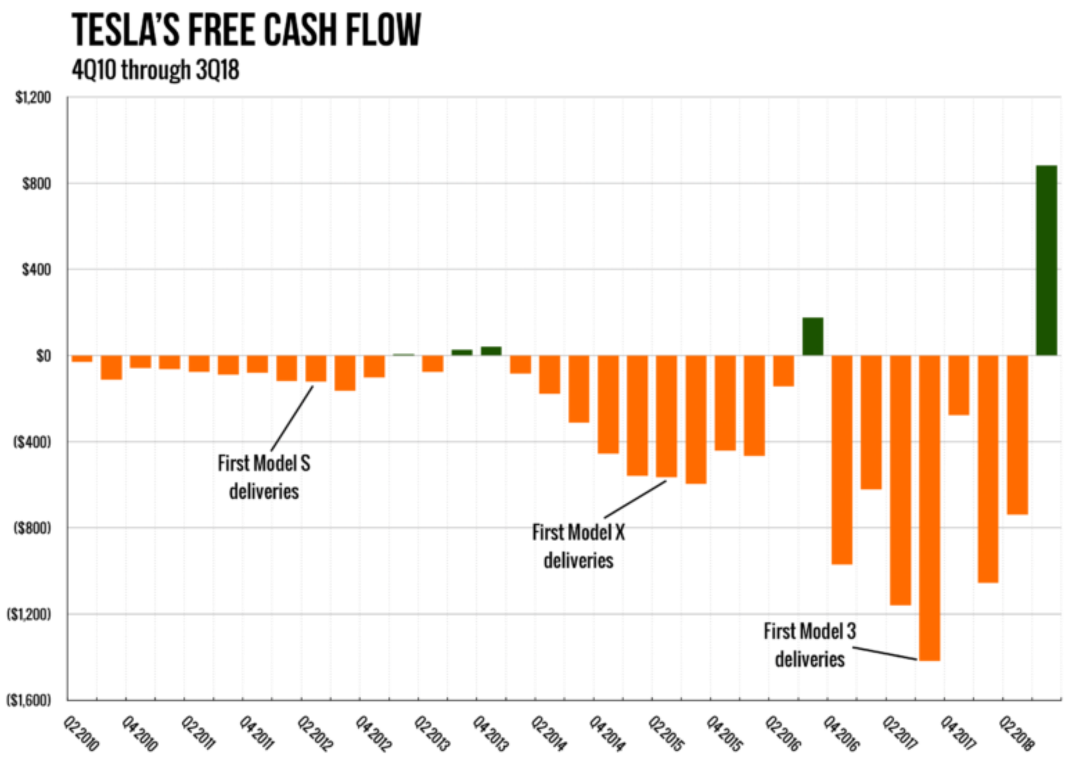

Figure 1. 4Q 2010 through 3Q 2018 Tesla's free cash flow (Lee, 2018)

From the creation of Tesla to the second quarter of 2019, the company has always had potential risks. There are three reasons and potential risks that could lead to Tesla's failure. One of the reasons why Tesla failed was that it had not made a profit from 2003 to the second quarter of 2019. In terms of numbers, Tesla's expenses grew faster than expected. As seen in Figure 1 above, Tesla's cash consumption rate has not decreased but has been increasing. Therefore, many people have been expecting Tesla to close down due to insufficient funds from the beginning, and such comments have never stopped. In May 2019, Tesla raised about $2.7 billion. Tesla CEO Elon Musk also realized that if the company continues to burn cash, Tesla will use up all the money it has in 10 months. Fortunately, Tesla's profitability in the third quarter made it succeed in the crisis. The report from MarketWatch shows that the adjusted earnings per share for the third quarter was $1.86, and the market expects a loss of $0.24. The net profit in the third quarter was $143 million. The market expects a loss of $257 million, compared with a profit of $311 million in the same period last year. (Claudia Assis, 2019)

Figure 2. 2010 to 2019 Tesla’s Stock Price (Yahoo Finance, 2019)

Tesla's second potential risk is that it is stock price volatility is enormous. It can be seen from Figure 2 that Tesla's stock price has been fluctuating from $145 to $383 over the past five years. This picture is like seeing the price of stocks up and down in the same way as the electrocardiogram of Tesla investors. For a large company like Tesla, it is infrequent for stock prices to have such significant volatility. A piece of good news is enough to make Tesla's stock rise by 10% to 30%, but a piece of bad news is enough to make Tesla's stock fell by the same percentage. According to recent data, Tesla's report for the first quarter of 2019 is not very good, so this quarter's release has caused Tesla's stock to fall by 32.2% from $264 on April 23rd to

$179 on June 3rd. Tesla's stock is also the one that many people like to sell short in the historic market. There are two reasons why Tesla's stock is so unstable and people want to sell it short. First, it is complicated to estimate the value of a fast-growing company. When Facebook and Amazon had not yet become mainstream industries, stock appraisers did not have many expectations for the industry of such companies. Because there is no historical record to compare new industrial companies, according to the historical cost principle, stock valuers can only be priced from a conservative and objective perspective. According to CNN, the 30 analysts offering 12-month price forecasts for Tesla, Inc. have a median target of $283.50, with a high estimate of $949.00 and a low estimate of $160.00. (CNN, 2019)

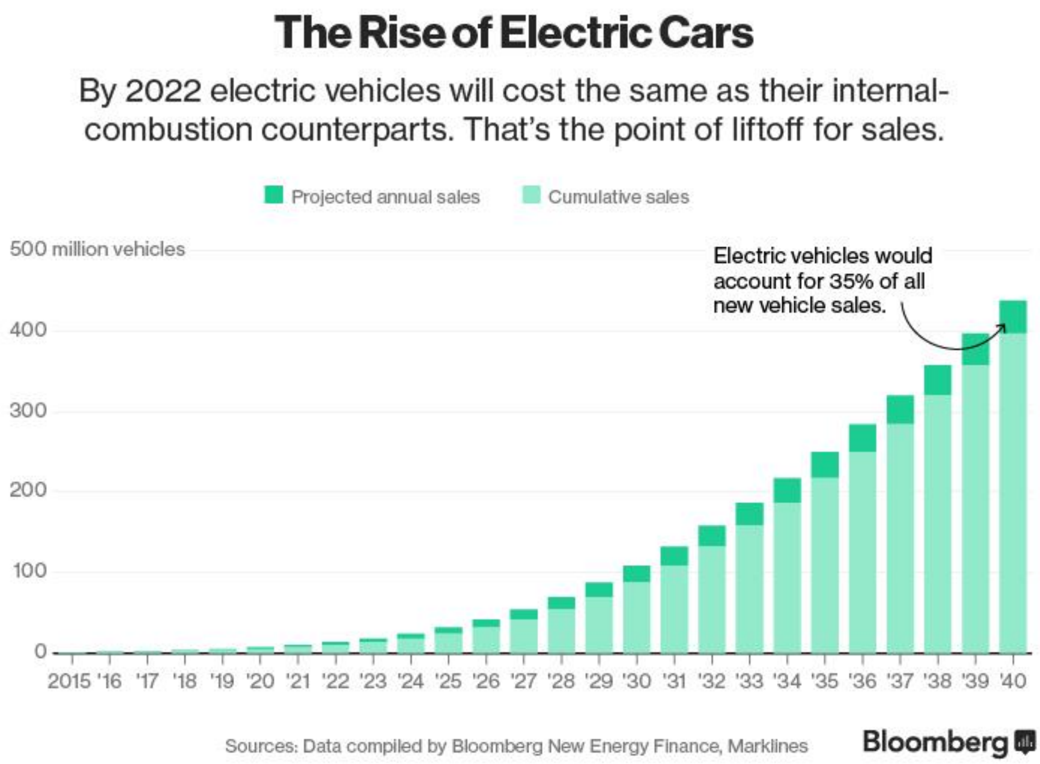

Figure 3. 2015 to 2040 The Rise of Electric Cars (Tom Randall, 2016)

The third potential risk is that the market demand for electric vehicles may not be as good as expected. As shown in Figure 3, according to information by Bloomberg, the demand for electric vehicles in the global market will increase significantly in the future. Still, it does not entirely occupy a leading position. Many new drivers in China and India will continue to choose gasoline and diesel. Rising oil demand from developing countries could outweigh the impact of electric cars, especially if crude prices fall to $20 a barrel and stay there (Tom Randall, 2016).

The present paper provides investors with financial analysis and asks two questions. In the face of three significant risks, what is the reason for Tesla's stock to rise in the future? What are the reasons behind Tesla’s stock surge after the financial report released in October 2019?

CHAPTER 2

The Background of Tesla, Inc.

Tesla is a technology company founded by Martin Eberhard and Marc Tarpenning in 2003. The idea for the company came from a group of engineers in Silicon Valley who aimed to reduce the world's dependence on fossil fuel vehicles by developing electric cars. Tesla currently designs, manufactures, and markets electric vehicles, as well as making components for other companies, including Toyota and Daimler. In 2004, Elon Musk decided to invest in the company. The company launched its initial public offering in 2007, raising more than $227 million. The company was rated the best performing company on the NASDAQ index in 2014.

The company's strategy of choosing direct customer sales, with its stores and service centers, is a departure from the established dealer model that now dominates the U.S. auto market. In 2008, the company introduced its first all-electric car, called the Tesla Roadster. With this launch, Tesla received a lot of attention, selling more than 2,400 Roadster cars in more than 30 countries. In 2010, the company went public and raised $226.1 million. Tesla is the first American carmaker to go public since Ford in 1956. In 2012, the company introduced its first premium electric sedan, called the Model S. Three years after the Model S, it launched the Model X. In 2017, Tesla unveiled its cheapest Model, Model 3. As Tesla gets bigger, the Model Y, Semi, and the new 2020 Roadster are already taking pre-orders. Tesla used the internet to create an online buying environment for clients. Online automatic buying has been proved by many people to be convenient and reliable.

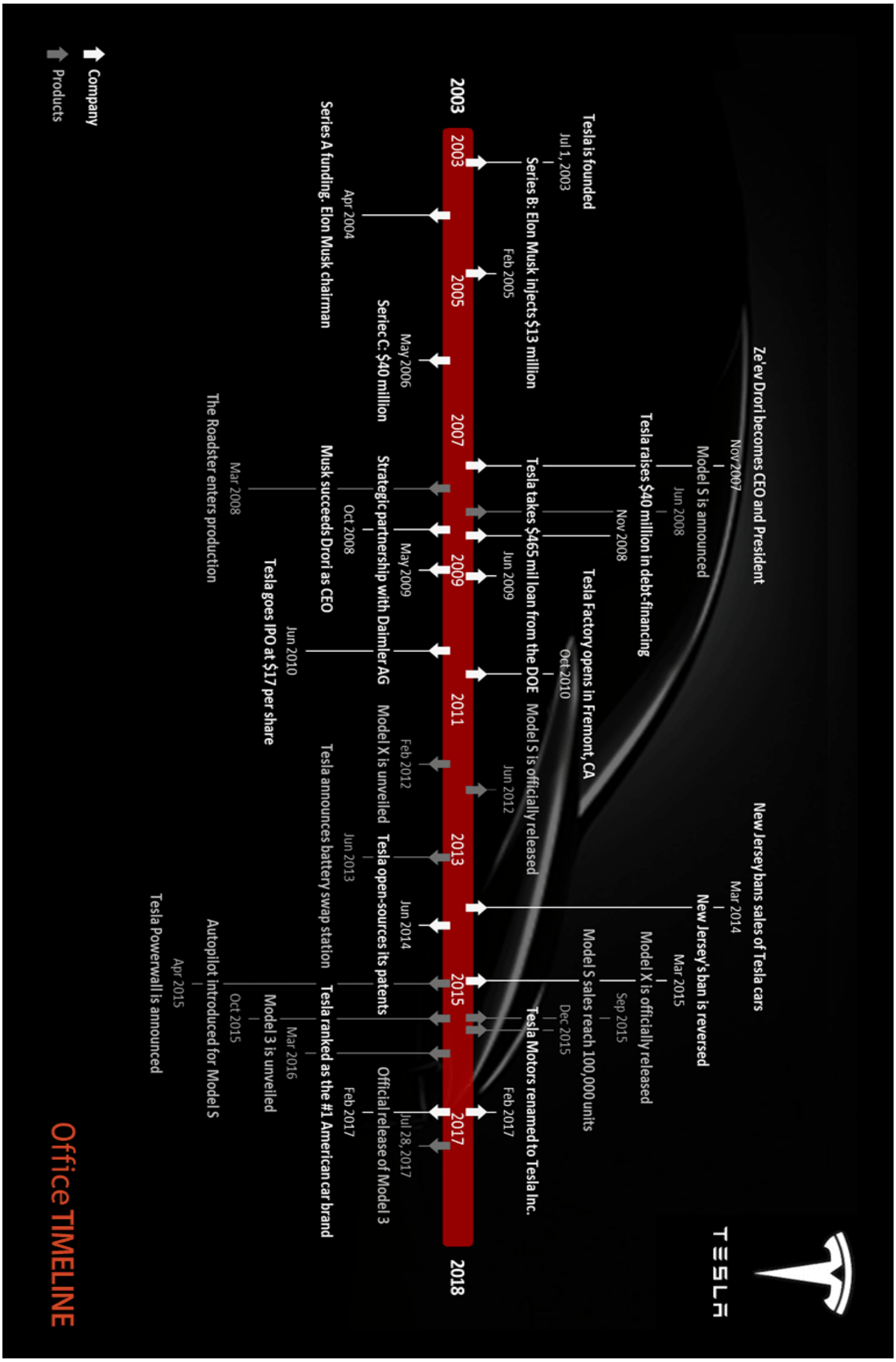

Figure 4. Tesla's Timeline

CHAPTER 3

External Analysis of Tesla: PESTEL and Porter's Five Forces Analysis

3.1. PESTEL Analysis

Political Factors

Government entities are one of the major social forces that influence industry and commerce. Trade policies may limit industry performance and company revenues. Three political externalities are critical to Tesla and the energy solutions industry:

- Government incentives for electric cars,
- New global trade agreements,
- Political stability in most major markets.

Government incentives could be an opportunity to help Tesla to strengthen its financial performance. This external factor is directly related to the company's operations and products to minimize carbon emissions. In other words, the political stability of significant markets makes the remote or macro-environment favorable to Tesla's generic competitive strategy and intensive growth strategy, including market penetration (Gillespie 2007).

Economic Factors

Market growth, trade levels, currencies, and other variables can all affect the car business. The growth rate of the solar market determines the growth opportunities for the company's solar panel business. Tesla's acquisition of SolarCity could be a solution for the future. Three external economic factors are affecting the car market that Tesla needs to address:

- Reduce battery cost,
- Reduce the cost of renewable energy,
- Economic stability.

Tesla's business performance benefited from lower battery costs. Reducing the cost of renewable energy could make Tesla's products more attractive. As renewable energy solutions become more popular, Tesla's business will improve. But economic stability problems threaten companies' financial performance, especially in Europe and Asia (Housing Industry Association 2011).

Social and Cultural Factors

Social factors are related to economic and political factors. They include cultural attributes, emphasis on environmental awareness, and perception of green products. Most people are concerned about the environment, greenhouse gases, and carbon footprint. It has become a trend for consumers to buy environmentally friendly products. There are also three crucial social and external cultural factors in Tesla's business:

- The popularization of a low-carbon lifestyle,
- The preference for renewable energy is growing,
- Improving the distribution of wealth in developing markets.

Tesla has growth opportunities based on the growing popularity of low-carbon lifestyles and a choice for renewable energy. As people become more environmentally conscious, this trend increases the number of potential buyers of the company's relatively expensive cars (Murphey & Gause 1974).

Technological Factors

Using electric technology to power vehicles is the latest technology that many admire. However, a rapid charging system is still a technology that needs to be continuously improved and updated. Besides, material engineering technology also determines the efficiency and cost-effectiveness of the company's battery. Three factors will affect Tesla:

- Rate of change in high technology,
- Enhanced business automation,
- The increasing popularity of online mobile systems.

High technical change rates are opportunities and threats in this business analysis. High-interest rates offer Tesla a chance to enhance its product technology. But the same external factors threaten the rapid obsolescence of techniques used in companies' products. Tesla has growth opportunities through further automation of its business processes. Moreover, the growing popularity of online mobile systems should prompt the company to integrate them increasingly into its cars (Roper 2012).

Environmental Factors

Environmental factors include awareness of climate change. Growing concerns are affecting the way companies do business and the products they offer.

Ecological trends determine the availability of materials used in a company's production process. In this case, there are three reasons why external environmental factors affect Tesla:

- Climate change,
- Expanded environmental program,
- Waste treatment standards are continually improving.

Tesla has a chance to promote its electric cars based on concerns about climate change, and developed ecological applications, and improved waste standards. Tesla's electric vehicles, batteries, and solar panels are considered suitable to directly address these external factors related to business sustainability and environmentally friendly products. Tesla's products will have a massive opportunity for growth in terms of environmental factors (Gillespie 2007).

Legal Factors

Laws and legal systems affect management decisions and business development. Tesla's marketing mix, or 4ps, is governed by law. Its human resource management and business partnerships are also legally bound. Tesla's strategy needs to include three external factors:

- Expanding international patent protection,
- Energy consumption regulation,
- Dealer sales regulations in the United States.

Because countries around the world attach great importance to patent protection, Tesla will also grow its business overseas. Tesla's electric vehicles and energy solutions products will have an opportunity based on the energy consumption regulations that the customer organization must observe. Moreover, they have the chance to grow through direct sales (Faroop 2019).

3.2. Porter's Five Forces Analysis

Level of Competitive Rivalry: High

The electric car industry is very competitive. Tesla has continued to grow over the past few years, posing a threat to other automakers such as BMW and Nissan. Together with Tesla, the two automakers have an 80% share of the electric car market. Because Tesla has achieved economies of scale through its Model S Model, Model X and Model 3, barriers to entry into the electric car market are high (Praem, 2014).

Bargaining power of buyers: Moderate

Buyers have moderate bargaining power. Tesla offers a differentiated product that combines luxury, prestige, performance, and environmental awareness, and has little competition in the electric car industry. Its competitors include the Toyota Prius, Nissan Leaf, and Chevrolet, which pose little threat to Tesla for now.

In many cases, the supply of substitutes is small, limiting customers' bargaining power with Tesla. In the current electric-car industry, consumers tend to be highly customized so that some fewer manufacturers can meet their needs (Praem, 2014).

Bargaining power of suppliers: Moderate

Tesla typically makes most of its equipment in-house, but to keep costs down, increasingly outsource parts to standardize with the industry. Battery components are also the most challenging and expensive parts of electric cars, designed and manufactured almost entirely by Tesla, reducing the power of suppliers. A large number of potential auto parts suppliers reduce the switching cost of auto companies, thus obtaining competitive transactions that reduce the strength of suppliers (Essays 2018).

Threat of Substitutes: Low

The threat of substitutes is minimal. The reason for the low threat is that any possible alternatives to the product would be hybrids, hydrogen, fuel cells, mass transit, and diesel. All of these may require the use of fossil fuels. As one of a company with zero-emission vehicles, Tesla has mastered the core battery technology of electric cars, so it has no competitive competitors in the short term. This makes alternatives less of a threat. People who care more about the environment and like sports cars may prefer Tesla products (Kissinger, 2019).

Threat of New entrants: Moderate

Because of the high cost of manufacturing and brand development, it is not easy for new entrants to establish themselves in the auto industry. For electric cars, consumers are more concerned about the battery life of electric vehicles. This is often the most difficult technical hurdle for new entrants to overcome. Although the barriers to electric cars are not low, most big car companies are heavily involved in the electric car industry, and the obstacles will not stop Nissan, Ford, BMW, and other established companies. These threats come from existing cars as well as technologies like Google and Apple. Like Tesla, they are tech companies that already have strong brand influence and want to enter the auto industry with high-tech electric and self-driving cars (Essays 2018).

Conclusion

The industry's capital intensity limits the threat of new entrants and forces players to reach critical mass. Buyers and suppliers have limited bargaining power because they are highly dependent on the industry. Fierce competition is the most important factor that restricts the profitability of the industry.

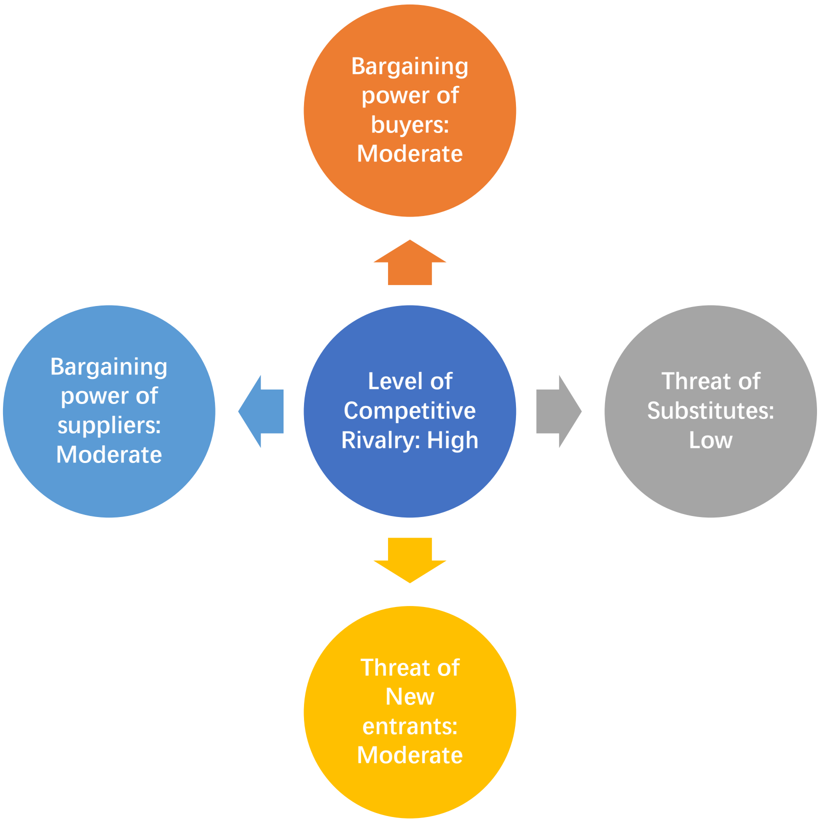

Figure 5. Porter's Five Forces Analysis

CHAPTER 4

Tesla, Inc.'s SWOT Analysis

Strengths

- Sustainable innovation - steady free cash flow, using Tesla stores to replace dealerships and popular with the younger generation who provide free advertising via social media.
- Strong brand awareness - Tesla is a recognized brand. The brand has gained fame and global recognition through its products and services. Its differentiated business model has won widespread publicity and attention. Despite some of the negative publicity surrounding Tesla's battery burns, the Internet has been abuzz with discussion about the brand and its products (Pratap 2019).
- Strong sales growth - Tesla has been proliferating over the past few years. Tesla sold 321 cars in the third quarter of 2012 and 83,500 in the third quarter of 2018, a period of 25,912% sales growth (Fosse, 2019).
- Battery life - Tesla has long been known for its efficient battery life. In the current automobile industry, the range of the Tesla battery pack occupies a monopoly position.
- Technology - Tesla has always been associated with cutting-edge technology, and many companies have adopted technologies introduced by Tesla. Some examples of top brands using Tesla technology include Daimler, which uses Tesla batteries; Mercedes is the user of the Tesla power system.

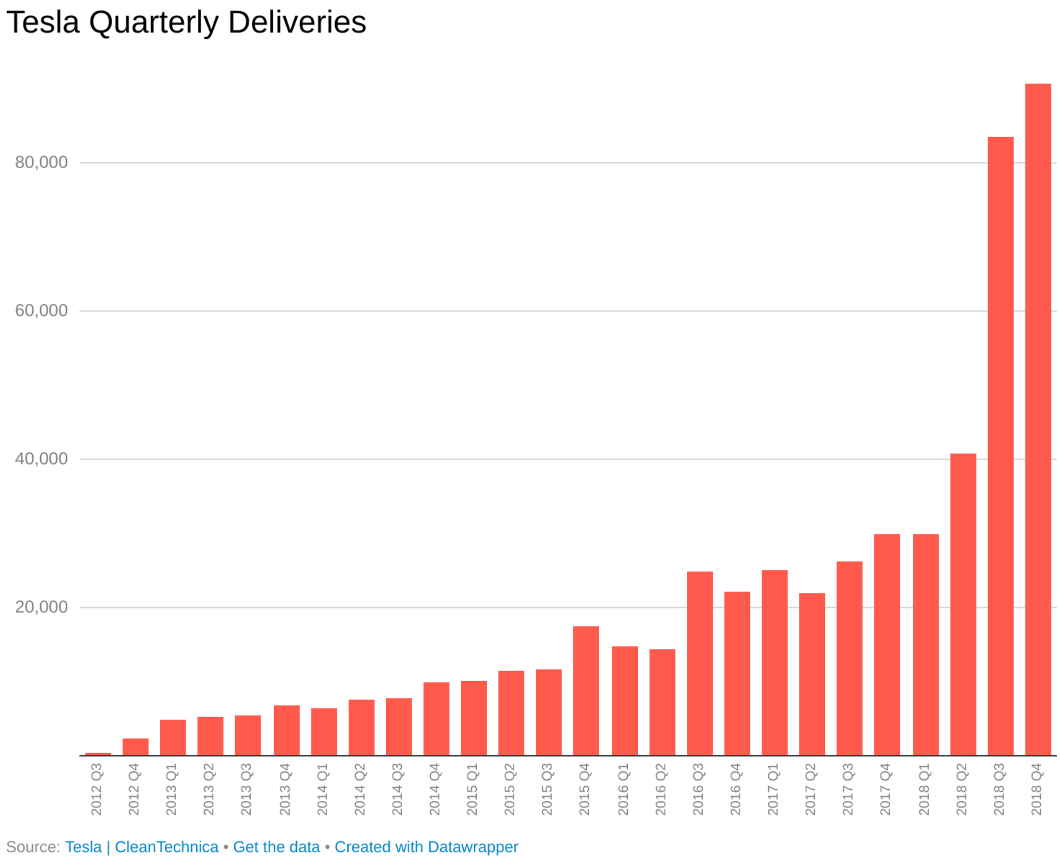

Figure 6. Tesla's Sales Growth

Weaknesses

- High debt burden - the company has a relatively high debt burden. At one point, Tesla had nearly $2.5 billion in long-term debt and capital leases on its balance sheet, about 72 percent of the total. That compares with just $1.4 billion in cash. Interest payments on the debt are substantial and are likely to continue to cut yields. If a company is unable to meet its debt obligations due to insufficient cash flow, then it must reduce investments and capital expenditures, which may hinder future growth (Pratap 2019).

- High production and operating costs -- Tesla's production and operating costs are high. Tesla has managed to cut manufacturing costs by increasing the creation of the

Model S and Model X. However, research and development and laws and regulations can be expensive, and compliance with these laws can lead to increased costs.

- Limited market presence - Tesla's market share is limited. The company derives most of its revenue from the United States and has few operations in China and developing countries. This internal strategic factor is a weakness that limits business growth based on the rapid economic development of overseas markets (Kissinger 2018).
- Poor customer awareness - Tesla makes very professional products because they are sustainable and, therefore, critical to the environment. But these products are highly futuristic, and most customers are still unsure whether they need to invest in these expensive technologies.
- Capacity issues - Tesla makes cars at only one plant in Fremont, California. The plant has a capacity of 500,000 vehicles, and the company's maximum size is limited to that number, making it challenging to target higher volumes (Pratap 2019).

Opportunities

- Global expansion - this opportunity is based on significant economic growth in countries such as Pakistan, where Tesla has little or no market share. Tesla could work with local dealers in Pakistan to target the young and affluent economic class.
- Autonomous driving technology - In the past ten years, the automobile industry has made significant progress driven by technological changes. Autonomous driving is currently an important area, and most leading car brands are still looking for

breakthroughs in autonomous driving technology. Tesla's autonomous driving technology has gained enough reputation for its safety and convenience. Autonomous driving technology has great potential and still has a lot of room for development. Autonomous driving is an essential direction for the development of automotive technology in the future. Over time, cars equipped with autopilots have become more and more popular. Tesla's autopilot has many features, but also has some restrictions, mainly to prevent driver abuse and carelessness (Pratap 2019).

- Asian markets - China is the largest automotive market. However, despite the U.S. accounting for nearly 70% of the brand's 2018 revenue, China's share is still below 10%. Given Tesla's ability to improve its dealer network in China, it may find a more extensive customer base in the world's second-largest economy. The Asian market is one of the fastest-growing markets in the world. Tesla has more opportunities in China. In addition to automobiles, there are also large markets for its other products (Pratap 2019).
- Government support - Tesla focuses on helping the world transition to a better and more energy-efficient world. Tesla can get enough support from the government to promote energy-efficient and environmentally friendly cars.
- Business diversification - Tesla successfully transformed itself in early 2019. It removed the word “Motors” from the company's registered name and replaced it with the word “Tesla, Inc.” After Tesla bought SolarCity, it's announced in October that it would go ahead and buy DeepScale, a technology company. Diversified channels will

provide more opportunities for Tesla (Wikipedia 2019).

Threats

- Aggressive competition - Tesla faces fierce competition from luxury and environmental brands. In addition to luxury brands such as Audi and Porsche, brands that make environmentally friendly cars, such as the NISSAN Leaf and Chevrolet Bolt, present significant competitive threats (Kissinger 2018).
- Legal and regulatory issues - legal and regulatory issues can be expensive transactions and can lead to increased costs. When expanding to other countries/regions, the business must comply with the laws and political regulations in these markets.
- Material price volatility - material price volatility is another threat. The price fluctuation and general rise of lithium materials used in the company's energy storage products will significantly increase the company's production costs (Yahoo Finance 2019).
- Poor infrastructure and lack of charging stations - in developing counties, the poor infrastructure and lack of charging stations can negatively affect demand for environmentally-friendly but expensive electric vehicles. Moreover, importing spare parts would lead to high maintenance costs, thus hindering a potential market (Kissinger 2018).
- Funding Production - Funding control of Tesla has long been a significant concern for investors. The talk that Tesla may not be able to fund its ambitious

production growth in the coming months has never stopped. A more effective way of controlling money would make Tesla's future path a little easier.

Recommendations

Tesla has the strength to keep the business thriving for years to come. However, solving problems to remain competitive and profitable is very important. Tesla should continue to strengthen its competitiveness and achieve a competitive advantage through innovation and increasing market share. In terms of change, Tesla can increase its investment in R&D to exceed the speed of innovation of energy storage of competitors. Furthermore, Tesla must improve its transnational operations. New facilities and sales operations in many high-potential developing countries can drive business growth and meet Tesla's corporate mission and vision. Tesla Should:

- Expand operations in foreign markets to take advantage of global growth in the renewable energy industry,
- Continue or increase investment in product innovation,
- Diversify the supply chain to reduce supplier risk.

CHAPTER 5

Tesla, Inc.'s Financial Analysis

In order to understand Tesla's financial position and forecast cash flow, assessing historical developments and performance is critical. By analyzing previous financial statements, one can see how Tesla creates value and how the company performs relative to its peers. As Figure 2 shows, Tesla's stock price has been highly volatile since 2013. Revenue grew by about 500% from 2012 to 2013, and the stock rose from $33.87 in 2012 to $150.43 in 2013. In 2016, just three years later, it's stock price rose from $213.69 in 2016 to $311.35 in 2017, doubling its revenue growth liquid. Based on historical performance to estimate future cash flow, Tesla has mature manufacturers in mature industries. Therefore, I believe that growth can be predicted and verified by analyzing Tesla's historical development.

5.1. The Analytical Income Statement

Table 1

Tesla's Income Statements Common Size

Income Statements	2019 Q3	2018	2017	2016	2015	2014	2013	2012
Net sales	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Operating costs	81.10%	81.17%	81.10%	77.15%	77.18%	72.43%	77.34%	92.72%
Gross profit	18.90%	18.83%	18.90%	22.85%	22.82%	27.57%	22.66%	7.28%
EBITDA	14.15%	7.27%	-0.87%	5.71%	-18.70%	-5.75%	-1.91%	-95.78%
Depreciation and amortization	8.42%	8.86%	13.91%	13.53%	10.44%	7.25%	5.27%	6.98%
EBIT	5.73%	-1.59%	-14.78%	-7.82%	-29.15%	-13.00%	-7.18%	102.76%
Less interest	2.94%	3.09%	4.01%	2.84%	2.94%	3.15%	1.64%	0.06%
Earnings before taxes (EBT)	2.79%	-4.68%	-18.79%	-10.66%	-32.09%	-16.15%	-8.82%	102.82%
Taxes (40%)	0.41%	0.27%	0.27%	0.38%	0.32%	0.29%	0.13%	0.03%
Net Income before preferred dividends	2.27%	-4.55%	-16.68%	-9.64%	-21.96%	-9.19%	-3.68%	-95.88%

Revenue is an important measure of performance and shows the results of a company's core activities without considering financing options. The financial statements of common size and percentage change make it easier to compare, analyze the company over time and compare it to competitors. The applicable percentages on the income statement and balance sheet may help investors spot trends.

In 2016, Tesla's operating costs was 77.15%, while the latest 2018 annual report indicated that Tesla's operating costs increased slightly to 81.17% of sales. Since 2012, Tesla's operating cost has been reduced from 92.72% to about 77%, and it has maintained this proportion steadily. The percentage of annual operating costs for sales has been dramatically reduced. Although the gross profit of Tesla has decreased in the past three years, the gross profit of Tesla has maintained a relatively stable proportion since the substantial increase since 2012. As of 2018, 2017, and 2016, Tesla's gross profit accounted for 18.83%, 18.90%, and 22.85% of sales, respectively. Tesla's surprisingly strong third-quarter included encouraging signs. But there's one thing that's different from the usual suspects: the car's gross margin (Morris, 2019).

Tesla's auto margins are sometimes used as a measure of whether the company can meet its long-term goal of mass-producing electric cars profitably. That is especially true now since it sells far more mid-range Model 3 sedans than its more profitable luxury cars. These stunning reversals come even though actual deliveries are growing by only 2% a quarter, and overall prices per vehicle are lower. For the first time since 2012, Tesla reported an 8% year-on-year drop in quarterly revenue to

$6.3 billion from $6.8 billion in the same period in 2018, Bloomberg reported (Morris, 2019).

5.2. The Analytical Balance Sheet

Table 2

Tesla's Balance Sheets Common Size

Balance Sheets	2019 Q3	2018	2017	2016	2015	2014	2013	2012
Assets								
Cash and equivalents	16.28%	12.39%	11.75%	14.97%	14.79%	32.58%	35.00%	18.12%
Short-term investments	0.71%	0.65%	0.54%	0.47%	0.28%	0.31%	0.12%	1.71%
Accounts receivable	3.44%	3.19%	1.80%	2.20%	2.09%	3.87%	2.03%	2.41%
Inventories	2.01%	10.47%	7.90%	9.12%	15.79%	16.30%	14.08%	24.10%
Total Current Assets	33.36%	27.93%	22.93%	27.62%	34.50%	54.68%	52.38%	47.10%
Net plant and equipment	60.51%	66.21%	71.51%	66.35%	64.19%	44.38%	46.38%	50.47%
Total Assets	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Liabilities and equity								
Accounts payable	10.57%	11.45%	8.34%	8.21%	16.55%	17.90%	17.06%	30.80%
Notes payable	12.75%	15.11%	12.26%	12.16%	-0.47%	1.45%	3.93%	-2.70%
Accruals	7.61%	7.04%	6.04%	5.34%	3.97%	2.96%	2.82%	2.70%
Total current liabilities	79.02%	80.64%	81.73%	75.56%	86.54%	84.41%	72.40%	88.81%
Long-term bonds	48.08%	47.04%	55.08%	49.85%	66.50%	62.11%	48.59%	58.01%
Total common equity	18.42%	16.55%	14.79%	20.97%	13.46%	15.59%	27.60%	11.19%
Total liabilities and equity	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

To determine a company's profitability, it is necessary to restructure the official balance sheet to identify the two drivers of earnings: operating activities and financial activities.

As of December 31, 2018, cash and equivalents of the percentage of the total assets of 12.39%. When the third quarter of 2019 release, cash and equivalents of the percentage of the total assets of up to 16.28%. Notably, Tesla has been using its assets to generate more and more cash, and inventory controls on total assets appear to have improved in October. As noted above, Tesla has been optimizing its supply chain to improve inventory control. Net plant and equipment as a percentage of total assets also declined over the past three years from the previous year; The new Tesla plant in Fremont offers a decent return on investment and manufacturing equipment to meet the growing demand for Tesla cars in the future. As of December 31, 2018, accounts payable was 11.45% of total liabilities and equity, up from 8.34% in 2017 and 8.21% in 2016. It is essential to recognize that accounts payable on total liabilities and equity have been negative over the past year. According to Bloomberg, Tesla has used a more significant percentage of convertible debt in the past few years to finance new factories in China (Hull & Recht 2018).

5.3. Ratio Analysis

Table 3

Tesla's Ratios

LIQUIDITY RATIOS					**Industry**
	2019 Q3	**2018**	**2017**	**2016**	**Average**
Liquidity ratios					
Current Ratio	**1.08**	**0.83**	**0.86**	**1.07**	**1.70**
Quick Ratio	**1.01**	**0.52**	**0.56**	**0.72**	**1.29**
ASSET MANAGEMENT RATIOS					**Industry**
	2019 Q3	**2018**	**2017**	**2016**	**Average**
Asset Management ratios					
Inventory Turnover	**9.55**	**6.89**	**5.19**	**3.39**	**11.63**
Days Sales Outstanding	**65.3**	**16.1**	**16.00**	**26.03**	**11.92**
Fixed Asset Turnover	**0.32**	**1.09**	**0.57**	**0.47**	**0.00**
Total Asset Turnover	**0.19**	**0.72**	**0.41**	**0.31**	**0.77**
DEBT MANAGEMENT RATIOS					**Industry**
	2019 Q3	**2018**	**2017**	**2016**	**Average**
Debt Management ratios					
Debt Ratio	**79.02%**	**80.64%**	**81.73%**	**75.56%**	**0.00%**
Debt-to-Equity Ratio	**3.77**	**4.17**	**4.47**	**3.09**	**0.72**
Market Debt Ratio	**99.75%**	**99.76%**	**99.78%**	**99.82%**	**0.00**
Times Interest Earned	**1.95**	**-0.52**	**-3.69**	**-2.75**	**0.00**
PROFITABILITY RATIOS					**Industry**
	2019 Q3	**2018**	**2017**	**2016**	**Average**
Profitability ratios					
Profit Margin	**2.27%**	**-4.55%**	**-16.68%**	**-9.64%**	**9.34%**
Basic Earning Power	**1.10%**	**-1.15%**	**-6.06%**	**-2.42%**	**0.00%**
Return on Assets	**0.44%**	**-3.28%**	**-6.84%**	**-2.98%**	**6.88%**
Return on Equity	**2.37%**	**-19.83%**	**-46.29%**	**-14.20%**	**11.82%**
MARKET VALUE RATIOS					**Industry**
	2019 Q3	**2018**	**2017**	**2016**	**Average**
Market Value ratios					
Price-to-Earnings Ratio	**46.26%**	**-5.83%**	**-2.64%**	**-4.56%**	**1.79%**
Price-to-Cash Flow Ratio	**0.10**	**0.06**	**-0.16**	**0.11**	**0**
Price-to-EBITDA	**0.02**	**0.01**	**-0.15**	**0.03**	**2.54**
Market-to-Book Ratio	**1.10%**	**1.16%**	**1.22%**	**0.65%**	**0.00%**

The current ratio decreased by 0.0292 from 2017 to 2018, indicating a decrease in current assets and the company's ability to pay short - and long-term debts. The quick ratio is an indicator of a company's short-term liquidity and measures its ability to meet short-term obligations with its most liquid assets. Compared to 2017, the ratio declined by 0.0443 in 2018, indicating that there are fewer current assets available to cover every dollar of short-term debt. Therefore the company's liquidity position has deteriorated. On the other hand, over the past three years, the company's long-term debt has been in the same range as its total assets. Although long-term debt increased significantly in 2018, it has been balanced by the purchase of capital assets to generate long-term benefits.

The company's Total Debt to Equity Ratio fell from 4.47 in 2017 to 4.17 in 2018. Compared with 2017 and 2016, the company's Gross Profit Margin Ratio increased by 12.13% in 2018. This indicates reduced operating costs and increased efficiency while increasing the selling price of the product. The return on assets ratio has been negative for the company over the past three years due to net losses. Similarly, the company's return on equity over the past three years has been negative because of net losses. In 2018, the company returned a - $0.3282 for every $1 invested.

For the past three years, the company's net profit margin has been generating a net loss. The percentage margin of net loss in 2018 was 12.13 percent lower than in 2017. In terms of cash return returns, the company has maintained substantial ending cash and bank balance, which is why it has a favorable ratio — finally, cashback assets and cashback equity. Neither ratio shows satisfactory results because the company has been losing money and, therefore, has not received any cash from operating activities.

As shown in Table 3, the three-year return on equity is explained by a negative net profit margin. Poor performance and reduced shareholder returns in 2018 due to insufficient sales, high operating costs, and large amounts of debt. In the third quarter of 2019, one can see a net improvement in performance. The net profit margin increased to 2.27%. In addition, profits generated by assets are higher than in 2018. The reduction in financial leverage will also help improve performance in 2019. Finally, in October 2019, the situation improved, and the shareholder made a profit of 2.37%, which was the first profit of Tesla since its establishment.

5.4. Analysis of Conclusion

Tesla, Inc. is one of the most famous automotive companies in the world. The company has been plagued by losses and hot news, attracting the attention of various stakeholders in the market. Consumers' shift toward environmentally-friendly car choices has led to a flurry of orders and the focus of other automakers to observe the company. Tesla's gross profits are good, but compared to the percentage of sales, it shows that the gross profit had dropped from 18.90% in 2017 to 18.83% in 2018. Higher maintenance and research and development costs and sales and general and administrative expenses have resulted in lower net profit margins. Tesla has never made a full-year profit. One reason that should not be ignored is interest expense.

Interest payments are far beyond their means. Long-term debt is high.

CHAPTER 6

Top 3 Reasons Why Tesla Has a Good Return

The stock market rewards companies that are living in a world that is constantly changing, rather than focusing on short-term profit streams. Tesla, as a startup company, has always been able to surprise people. At the beginning of this paper, Tesla's three potential risks were mentioned, and according to previous years' data reports, Tesla's ratios are indeed not optimistic. Even so, Tesla will still have a two to three times return on the next five to ten years.

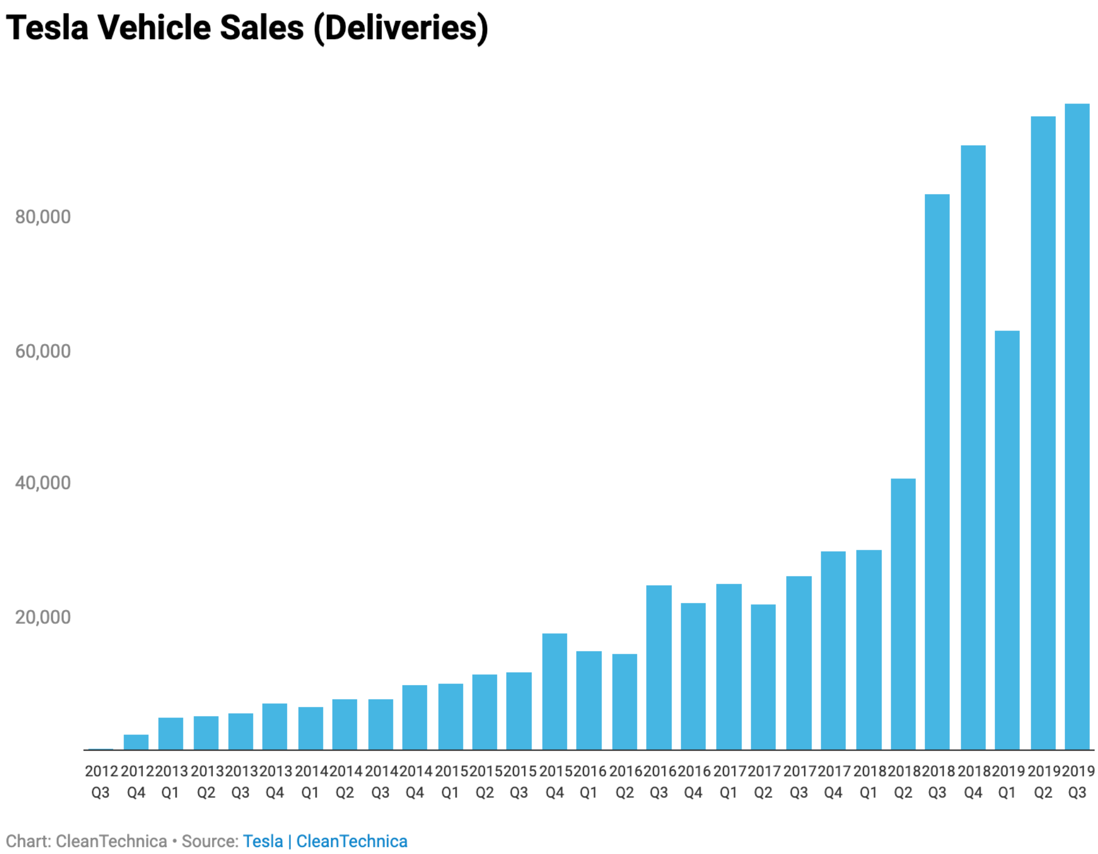

Figure 7. 2012 to 2019 Tesla Vehicle Sales

The first reason is Tesla's high sales growth. From the sales data, one can often see the sustainability of a company. Tesla's sales volume is growing dramatically every year. The stock price of a company represents the profitability and development potential of a company. Comparing Tesla's financial statements for the second quarter of 2018, Tesla sold 40,000 cars in the second quarter of 2018. In the second quarter of 2019, just a year later, Tesla sold a record-breaking 95,000 vehicles. The performance of the two years has increased by 140%, and this fantastic growth rate is not just happening once (Wagner 2019).

Tesla company is a very new company compared to other car brands. The company is only 16 years old. But since Tesla went public, Tesla's sales figures have grown at an amazing rate, and excellent sales figures are closely related to the second reason. The second reason for a good return is that Tesla's income has been growing. Even Tesla’s rate of increase in revenue before the release of the financial report for the third quarter of 2019 is staggering. Referring to Tesla's revenue for the past three years, Tesla's revenue in 2016 was $7 billion. In 2017, Tesla's revenue increased to $12 billion, and in 2018 Tesla's revenue reached $21 billion. The average income growth in the past three years is 75%. Comparing Apple's revenue growth over the same period, Apple's revenue growth from 2016 to 2018 is 6% to 15% (Levin 2017).

The third reason for a good return in the future development potential of Tesla. According to Bloomberg, today's electric cars account for 2% of new car sales. It is expected that the market and demand for electric vehicles will continue to increase in the future. In 2040, electric vehicles will account for 35% of new car sales, which will exceed today’s 17 times the market. Most importantly, Tesla will have no competitors for many years. According to figure 8, Tesla Model 3 completely dominates the market for electric cars and leaves all competitors behind. Long-established car brands like BMW and Audi are just entering the electronics market. Although the performance of Tesla is still a controversial topic, it has to be acknowledged that Tesla's technology in the electronic car market far exceeds other brands for three to five years (Stewart 2018).

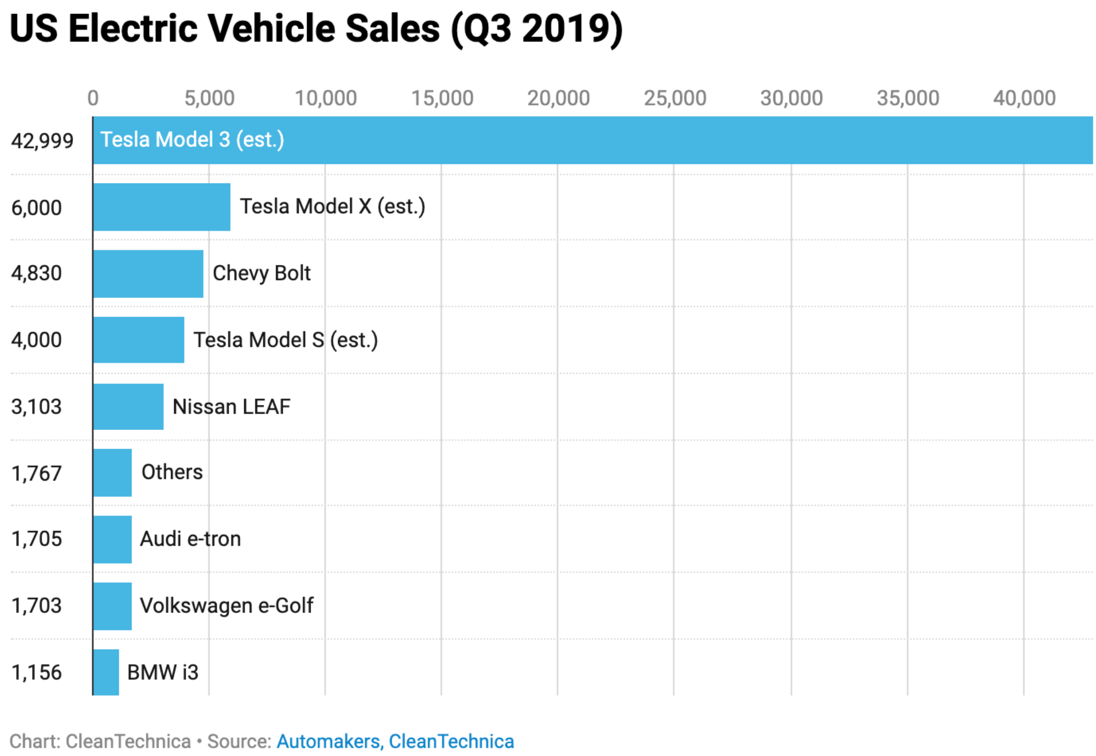

Figure 8. 2019 Q3 US Electric Vehicle Sales

Tesla's plans are also very attractive. First, the new factory has been built in China. Tesla's whereabouts can be seen in many big cities in China. Both Chinese citizens and the Chinese government also support Tesla in China. In China, the purchase of Tesla requires a lot of transportation and entry fees, but there are still a large number of people who go to import and purchase Tesla. The Chinese government also strongly supports Tesla to come to China because environmental pollution has become a severe problem in China's big cities. The arrival of Tesla can improve China's air problems. Tesla expects the annual output of the Chinese factory to reach 500,000 electronic vehicles. The Chinese market will also bring new high sales to Tesla. The best-selling Model 3 is priced at $44,000 in the US and $61,000 in China. When the Chinese market is fully developed, the actual benefits that can be given to Tesla could be an astronomical figure. In Tesla's plan, a fully automated driving taxi will be launched in the future, which will have a considerable impact on industries like Uber’s (Savaskan 2019).

Tesla's financial report in October 2019 gave everyone a big surprise. Tesla's share price rose from $220 to more than $350 in just over a month. Tesla announced in October that it had bought a technology company called DeepScale. DeepScale is a company that develops low-power processors that have technology that allows Tesla cars to use less energy to provide more accurate computer vision. This move by Tesla is to increase the speed at which it is developing fully autonomous driving. This performance report caused Tesla's stock to rise by 20% in one day. Wall Street initially expected Tesla to lose $0.42, but Tesla's bulletin's income was $1.86 (Randewich 2017). Tesla now proves that it can produce not only a large number of electric cars but also control costs. It can be seen that since 2016, Tesla's production has increased in Dafu every season, and at the same time, its income and cash control are getting better and better. According to the latest news from Tesla, it is expected that Model Y will be launched in the summer of 2020. Model Y is a 7-person electric car, and SUVs are more popular than vehicles in the US market. Therefore, what Tesla needs to do in the next year is to maintain Model 3 sales while enabling Model Y to be successfully launched (Kane 2019).

CHAPTER 7

Conclusion

Tesla has many characteristics of a growing company, with negative operating income and high investment. It also has many of the attributes of disruptive companies because it actively goes head-to-head with large and savvy industry peers to build a mass market for electric vehicles. In the past year, the company's revenue grew by about 82.51%, and its share price subsequently rose from $250 to $350.However, since 2010, Tesla has achieved negative results on various earnings indicators.

Macroeconomic factors will determine Tesla's growth, such as GDP growth, oil prices, and battery costs. Battery costs are a significant hurdle Tesla will have to overcome in its push to popularize electric cars. Tesla also depends on factors within the company's control, such as store expansion and charging networks. As a new player in a capital-intensive industry, Tesla will need to invest heavily in growth to reach critical mass.

In estimating future cash flows, it would be a good idea to focus on production capacity, capital investment, and battery costs. According to the analysis, these factors are the key drivers of value in the next few years. With the introduction of two new automotive platforms, growth will be achieved in two increments. By 2020, unit sales will be about 398,000, and EBITDA margins will be 14.5%. Current market prices also indicate that the market has priced much of the company's future profitability. Finally, according to the sensitivity analysis, the stock price is highly sensitive to the development of battery cost.

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APPENDIX

RATIO ANALYSIS (Section 3.1)

Input Data:

	2019 Q3	2018	2017	2016
Year-end common stock price	$359.52	$332.80	$311.35	$213.69
Year-end shares outstanding (in millions)	184.0	171.0	166.0	144.0
Tax rate	15%	-6%	-1%	-4%

Balance Sheets
(in millions of dollars)

Assets	2019 Q3	2018	2017	2016
Cash and equivalents	$5,338,000	$3,685,618	$3,367,914	$3,393,216
Restricted cash	$233,000	$192,551	$155,323	$105,519
Accounts receivable	$1,128,000	$949,022	$515,381	$499,142
Prepaid expenses and other current assets	$3,581,000	$365,671	$268,365	$194,465
Short-term marketable securities	$0	$0	$0	$0
Prepaid expenses and other current assets	$660,000	$3,113,446	$2,263,537	$2,067,454
Total current assets	$10,940,000	$8,306,308	$6,570,520	$6,259,796
Net plant and equipment	$19,845,000	$19,691,231	$20,491,616	$15,036,917
Other	$2,010,000	$1,742,075	$1,593,236	$1,367,363
Total assets	$32,795,000	$29,739,614	$28,655,372	$22,664,076
Liabilities and equity				
Accounts payable	$3,468,000	$3,404,451	$2,390,250	$1,860,341
Notes payable	$4,181,000	$4,493,432	$3,514,531	$2,756,636
Accruals	$2,497,000	$2,094,253	$1,731,366	$1,210,028
Total current liabilities	$10,146,000	$9,992,136	$7,636,147	$5,827,005
Total non-current liabilities	$15,767,000	$13,989,838	$15,784,637	$11,298,985
Noncontrolling interests	$0	$0	$0	$0
Total liabilities	$25,913,000	$23,981,974	$23,420,784	$17,125,990
Noncontrolling interests in subsidiaries	$842,000	$834,397	$997,346	$785,175
Common stock	$0	$173	$169	$161
Additional paid-in capital	$12,348,000	$10,249,120	$9,178,024	$7,773,727
Treasury stock, at cost	$0	$0	$0	$0
Retained earnings	-$6,188,000	-$5,317,832	-$4,974,299	-$2,997,237
Accumulated other comprehensive loss	-$120,000	-$8,218	$33,348	-$23,740
Total common equity	$6,040,000	$4,923,243	$4,237,242	$4,752,911
Total liabilities and equity	$32,795,000	$29,739,614	$28,655,372	$22,664,076

2015	2014	2013	2012	2011	2010
$240.01	$222.41	$150.43	$33.87	$28.56	$26.63
128.0	125.0	119.0	107.0	100.0	51.0
-1%	-2%	-1%	0%	0%	0%

2015	2014	2013	2012	2011	2010
$1,196,908	$1,905,713	$845,889	$201,890	$255,266	$99,558
$22,628	$17,947	$3,012	$19,094	$23,476	$73,597
$168,965	$226,604	$49,109	$26,842	$9,539	$6,710
$125,229	$94,718	$27,574	$8,438	$9,414	$10,839
$0	$0	$0	$0	$25,061	$0
$1,277,838	$953,675	$340,355	$268,504	$50,082	$45,182
$2,791,568	$3,198,657	$1,265,939	$524,768	$372,838	$235,886
$5,194,737	$2,596,011	$1,120,919	$562,300	$310,171	$122,599
$106,155	$54,583	$30,072	$27,122	$30,439	$27,597
$8,092,460	$5,849,251	$2,416,930	$1,114,190	$713,448	$386,082
$1,338,945	$1,046,830	$412,221	$343,180	$56,141	$28,951
-$38,222	$84,535	$95,100	-$30,088	$112,961	$44,859
$321,592	$173,052	$68,053	$30,088	$22,237	$11,755
$1,622,315	$1,304,417	$575,374	$343,180	$191,339	$85,565
$5,381,201	$3,633,124	$1,174,435	$646,310	$298,064	$93,469
$0	$0	$0	$0	$0	$0
$7,003,516	$4,937,541	$1,749,809	$989,490	$489,403	$179,034
$0	$0	$0	$0	$0	$0
$131	$126	$123	$115	$104	$95
$3,414,692	$2,345,266	$1,806,618	$1,190,191	$893,336	$621,935
$0	$0	$0	$0	$0	$0
-$2,322,323	-$1,433,682	-$1,139,620	-$1,065,606	-$669,392	-$414,982
-$3,556	$0	$0	$0	-$3	$0
$1,088,944	$911,710	$667,121	$124,700	$224,045	$207,048
$8,092,460	$5,849,251	$2,416,930	$1,114,190	$713,448	$386,082

Income Statements				
(in millions of dollars)				
	2019 Q3	2018	2017	2016
Total Revenue	$6,303,000	$21,461,268	$11,758,751	$7,000,132
Operating costs	$5,112,000	$17,419,247	$9,536,264	$5,400,875
Income from operating investments, net	$100,000	-$88,834	-$105,687	$119,802
General and administrative expense	-$596,000	-$2,834,491	-$2,476,500	-$1,432,189
Research and development expense, net	-$334,000	-$1,460,370	-$1,378,073	-$834,408
Depreciation and amortization	$530,851	$1,901,050	$1,636,003	$947,099
Loss on dispositions, net	$0	$0	$0	$0
Earnings before interest, taxes, depr. & amort. (EBITDA)	$891,851	$1,559,376	-$101,770	$399,561
Depreciation	$530,851	$1,901,050	$1,636,003	$947,099
Amortization	$0	$0	$0	$0
Depreciation and amortization	$530,851	$1,901,050	$1,636,003	$947,099
Earnings before interest and taxes (EBIT)	$361,000	-$341,674	-$1,737,773	-$547,538
Other income/(loss), net	$0	$0	$0	$0
Less interest	$185,000	$663,071	$471,259	$198,810
Earnings before taxes (EBT)	$176,000	-$1,004,745	-$2,209,032	-$746,348
Taxes (40%)	$26,000	$57,837	$31,546	$26,698
Net loss on disposal of discontinued operations	-$7,000	$86,491	$279,178	$98,132
Net income before preferred dividends	$143,000	-$976,091	-$1,961,400	-$674,914
Preferred dividends	$0	$0	$0	$0
Net income available to common stockholders	$143,000	-$976,091	-$1,961,400	-$674,914
Common dividends	$0	$0	$0	$0
Addition to retained earnings	$143,000	-$976,091	-$1,961,400	-$674,914

2015	2014	2013	2012	2011	2010
$4,046,024	$3,198,356	$2,013,496	$413,256	$204,242	$116,744
$3,122,521	$2,316,685	$1,557,234	$383,189	$142,647	$86,013
-$462,734	-$228,992	-$83,292	-$30,365	-$2,391	-$6,325
-$922,232	-$603,660	-$285,569	-$150,372	-$104,102	-$84,573
-$717,900	-$464,700	-$231,976	-$273,978	-$208,981	-$92,996
$422,590	$231,931	$106,083	$28,825	$16,919	$10,623
$0	$0	$0	$0	$0	$0
-$756,773	-$183,750	-$38,492	-$395,823	-$236,960	-$142,540
$422,590	$231,931	$106,083	$28,825	$16,919	$10,623
$0	$0	$0	$0	$0	$0
$422,590	$231,931	$106,083	$28,825	$16,919	$10,623
-$1,179,363	-$415,681	-$144,575	-$424,648	-$253,879	-$153,163
$0	$0	$0	$0	$0	$0
$118,851	$100,886	$32,934	$254	$43	$992
-$1,298,214	-$516,567	-$177,509	-$424,902	-$253,922	-$154,155
$13,039	$9,404	$2,588	$136	$489	$173
$422,590	$231,931	$106,083	$28,825	$0	$0
-$888,663	-$294,040	-$74,014	-$396,213	-$254,411	-$154,328
$0	$0	$0	$0	$0	$0
-$888,663	-$294,040	-$74,014	-$396,213	-$254,411	-$154,328
$0	$0	$0	$0	$0	$0
-$888,663	-$294,040	-$74,014	-$396,213	-$254,411	-$154,328

Common Size Statements

Balance Sheets	2019 Q3	2018	2017	2016
Assets				
Cash and equivalents	16.28%	12.39%	11.75%	14.97%
Short-term investments	0.71%	0.65%	0.54%	0.47%
Accounts receivable	3.44%	3.19%	1.80%	2.20%
Inventories	2.01%	10.47%	7.90%	9.12%
Total Current Assets	33.36%	27.93%	22.93%	27.62%
Net plant and equipment	60.51%	66.21%	71.51%	66.35%
Total Assets	100.00%	100.00%	100.00%	100.00%
Liabilities and equity				
Accounts payable	10.57%	11.45%	8.34%	8.21%
Notes payable	12.75%	15.11%	12.26%	12.16%
Accruals	7.61%	7.04%	6.04%	5.34%
Total current liabilities	79.02%	80.64%	81.73%	75.56%
Long-term bonds	48.08%	47.04%	55.08%	49.85%
Total common equity	18.42%	16.55%	14.79%	20.97%
Total liabilities and equity	100.00%	100.00%	100.00%	100.00%

Income Statements	2019 Q3	2018	2017	2016
Net sales	100.00%	100.00%	100.00%	100.00%
Operating costs	81.10%	81.17%	81.10%	77.15%
Gross profit	18.90%	18.83%	18.90%	22.85%
EBITDA	14.15%	7.27%	-0.87%	5.71%
Depreciation and amortization	8.42%	8.86%	13.91%	13.53%
EBIT	5.73%	-1.59%	-14.78%	-7.82%
Less interest	2.94%	3.09%	4.01%	2.84%
Earnings before taxes (EBT)	2.79%	-4.68%	-18.79%	-10.66%
Taxes (40%)	0.41%	0.27%	0.27%	0.38%
Net Income before preferred dividends	2.27%	-4.55%	-16.68%	-9.64%

2015	2014	2013	2012	2011	2010
14.79%	32.58%	35.00%	18.12%	35.78%	25.79%
0.28%	0.31%	0.12%	1.71%	3.29%	19.06%
2.09%	3.87%	2.03%	2.41%	1.34%	1.74%
15.79%	16.30%	14.08%	24.10%	7.02%	11.70%
34.50%	54.68%	52.38%	47.10%	52.26%	61.10%
64.19%	44.38%	46.38%	50.47%	43.47%	31.75%
100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
16.55%	17.90%	17.06%	30.80%	7.87%	7.50%
-0.47%	1.45%	3.93%	-2.70%	15.83%	11.62%
3.97%	2.96%	2.82%	2.70%	3.12%	3.04%
86.54%	84.41%	72.40%	88.81%	68.60%	46.37%
66.50%	62.11%	48.59%	58.01%	41.78%	24.21%
13.46%	15.59%	27.60%	11.19%	31.40%	53.63%
100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

2015	2014	2013	2012	2011	2010
100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
77.18%	72.43%	77.34%	92.72%	69.84%	73.68%
22.82%	27.57%	22.66%	7.28%	30.16%	26.32%
				-	-
-18.70%	-5.75%	-1.91%	-95.78%	116.02%	122.10%
10.44%	7.25%	5.27%	6.98%	8.28%	9.10%
			-	-	-
-29.15%	-13.00%	-7.18%	102.76%	124.30%	131.20%
2.94%	3.15%	1.64%	0.06%	0.02%	0.85%
			-	-	-
-32.09%	-16.15%	-8.82%	102.82%	124.32%	132.05%
0.32%	0.29%	0.13%	0.03%	0.24%	0.15%
				-	-
-21.96%	-9.19%	-3.68%	-95.88%	124.56%	132.19%

Percentage Change Analysis

Balance Sheets	2019 Q3	2018	2017	2016
Assets				
Cash and equivalents	44.83%	9.43%	-0.75%	183.50%
Short-term investments	21.01%	23.97%	47.20%	366.32%
Accounts receivable	18.86%	84.14%	3.25%	195.41%
Inventories	-78.80%	37.55%	9.48%	61.79%
Total Current Assets	31.71%	26.42%	4.96%	124.24%
Net plant and equipment	0.78%	-3.91%	36.28%	189.46%
Total Assets	10.27%	3.78%	26.44%	180.06%
Liabilities and equity				
Accounts payable	1.87%	42.43%	28.48%	38.94%
				-
Notes payable	-6.95%	27.85%	27.49%	7312.17%
Accruals	19.23%	20.96%	43.08%	276.26%
Total current liabilities	1.54%	30.85%	31.05%	259.18%
Long-term bonds	12.70%	-11.37%	39.70%	109.97%
Total common equity	22.68%	16.19%	-10.85%	336.47%
Total liabilities and equity	10.27%	3.78%	26.44%	180.06%

Income Statements	2019 Q3	2018	2017	2016
Total Revenue	-70.63%	82.51%	67.98%	73.01%
Operating costs	-70.65%	82.66%	76.57%	72.97%
		-	-	
EBITDA	-42.81%	1632.26%	125.47%	-152.80%
Depreciation and amortization	-72.08%	16.20%	72.74%	124.12%
EBIT	-205.66%	-80.34%	217.38%	-53.57%
Less interest	-72.10%	40.70%	137.04%	67.28%
Earnings before taxes (EBT)	-117.52%	-54.52%	195.98%	-42.51%
Taxes (40%)	-55.05%	83.34%	18.16%	104.75%
Net Income before preferred dividends	-114.65%	-50.23%	190.61%	-24.05%

2015	2014	2013	2012	2011	2010
-37.19%	125.29%	318.99%	-20.91%	156.40%	0.00%
26.08%	495.85%	-84.23%	-18.67%	-68.10%	0.00%
-25.44%	361.43%	82.96%	181.39%	42.16%	0.00%
33.99%	180.20%	26.76%	436.13%	10.85%	0.00%
-12.73%	152.67%	141.24%	40.75%	58.06%	0.00%
100.10%	131.60%	99.35%	81.29%	153.00%	0.00%
38.35%	142.01%	116.92%	56.17%	84.79%	0.00%
27.90%	153.95%	20.12%	511.28%	93.92%	0.00%
- 145.21%	-11.11%	-416.07%	- 126.64%	151.81%	0.00%
85.84%	154.29%	126.18%	35.31%	89.17%	0.00%
24.37%	126.71%	67.66%	79.36%	123.62%	0.00%
48.11%	209.35%	81.71%	116.84%	218.89%	0.00%
19.44%	36.66%	434.98%	-44.34%	8.21%	0.00%
38.35%	142.01%	116.92%	56.17%	84.79%	0.00%

2015	2014	2013	2012	2011	2010
26.50%	58.85%	387.23%	102.34%	74.95%	0.00%
34.78%	48.77%	306.39%	168.63%	65.84%	0.00%
311.85%	377.37%	-90.28%	67.04%	66.24%	0.00%
82.21%	118.63%	268.02%	70.37%	59.27%	0.00%
183.72%	187.52%	-65.95%	67.26%	65.76%	0.00%
17.81%	206.33%	12866.14%	490.70%	-95.67%	0.00%
151.32%	191.01%	-58.22%	67.34%	64.72%	0.00%
38.65%	263.37%	1802.94%	-72.19%	182.66%	0.00%
202.23%	297.28%	-81.32%	55.74%	64.85%	0.00%

Tesla's current debt ratio 81.10%

a) Weighted Average Cost of Capital

$= E/(D+E)*(R_E)+D/(D+E)*R_D*(1\text{-tax})=$ 9.87%

b) Weighted Average Cost of Capital

$= E/(D+E)*(R_E)+D/(D+E)*R_D*(1\text{-tax})=$ 1.46%

c)

Debt Ratio	Equity Ratio	r_E	r_D	**WACC**
0%	100%	1.41%	0%	1.41%
10%	90%	1.40%	3.06%	1.45%
20%	80%	1.38%	12.31%	2.71%
30%	70%	1.37%	12.31%	3.36%
40%	60%	1.35%	12.31%	4.01%
50%	50%	1.32%	24.00%	8.46%
60%	40%	1.27%	24.00%	9.87%
70%	30%	1.20%	24.00%	11.28%
80%	20%	1.05%	24.00%	12.69%
90%	10%	0.62%	24.00%	14.10%
99%	1%	7.28%	24.00%	15.37%

d)

WACC is the lowest around 1.41%.

The optimal debt ratio is around 0%.